NAME: S.MAGALAKSHMI

REG No: 192524144

SUBJECT: Fundamental Of Data Science

DATE: 30/9/2026

CO1 - ASSESSMENT TOOL 1

SHORT ANSWER TEST

1. A College wants to identify Students who may fail in the final exam. How can Data Science help?

Data Science can understand or can analyze Students marks, attendance, assignment Scores and previous performance. Using this data, it can predict which Student are at risk of failings, allowings teachers to provide earlys Support and guidance.

2. A hospital has patient records for many years. How can data Science be applied?

Data Science can analyze historical patient records, Symptoms, treatments and medical test result to identify disease patterns and pridict health risks. It can be help doctors make better decisions and improve patient care.

3. An Online Shoppings Websites wants to recommend products to customers. Which use of Data Science is applied here?

A Recommendation System is used. Data Science analyzes customers past purchases, Searches, views and preferences to recommend products that they may be intrusted in.

4. A bank wants to detect Suspicious transactions. How can Data Science be useful?

Data Science can analyze transaction amount, location, time, frequency and customer behaviour to identify unusual patterns. Transactions that

differ significantly from normal behaviour can be flagged as potentially fraudulent.

5. A Company wants to reduce customer loss. How can data Science Support this?

Data Science can be used for customer churn prediction. It analyzes customer behaviour, purchase history, complaints, and usage patterns to identify ~~not~~ customers who may leave, So the company can take Suitable actions to retain them.

6. A School wants to improve Students performance. Mention one benefit of Data Science in this case.

Data Science can identify Students Strengths, weaknesses and learnings Patterns. This help teachers to provide personalized Support and improve overall Student pefomance.

7. A Supermarket wants to know which products Sell more during festivals. How can Data Science help?

Data Science can analyze past Sales data during different festivals to identify products with high demand. The Supermarket can use these insights to maintain Sufficient Stock and plan future Sales.

8. A transport department wants to reduce traffic Congestion. How can Data Science be applied?

Data Science can analyze traffic Volume, vehicle environment, road Conditions, and travel patterns. It can identify Congested areas and help the department optimize traffic Signal, routes and transportation planning.

9. A farmer wants to improve crop yield. Give One use of Data Science.

Data Science can analyze Soil Conditions, weather, rainfall, temperature and crop data to recommand Suitable crops and farming practice. This can help farmers improve crop productivity, and make better decisions.

Company has data in Excel tables, Customer reviews, images and videos which facet of data is shown here?

The facet of data shown is Variety. Variety refers to the presence of different types and formats of data, such as structured data in Excel, text in reviews, images and videos.

11. A Weather app update temperature every minute. Which facet of data is involved?

The facet is velocity. Velocity refers to the presence of different types and formats of data, such as speed at which data is generated, collected, and processed. Since the temperature data is updated every minute, it demonstrates high data velocity.

12. A University stores student details in rows and columns. What type of data structure this?

This is structured Data. Structured data is organized in a fixed format, usually in rows and columns, making it easy to store, search, and analyze.

13. Customer Comments on social media are collected for analysis. What type of data is this?

Customer Comments on social media are generally unstructured data because they are free form text without a fixed structure such as rows and columns. Data Science can analyze these comments to understand customer opinions and sentiment.

14. A Company collects a very large amount of daily user activity data from its mobile app. Which Big Data characteristic is applied?

The Big Data characteristic is Volume. Volume refers to the large quantity of data generated and collected. The company collects a very large amount of user activity data every day.

15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?

The dataset has data quality issues involving completness and accuracy.

Missing age values represent a Completeness problem, incorrect phonenumber - represent accuracy problem

16. A business wants to convert customer data into useful Sales decisions. Which Big Data characteristic is important?

The important Big Data characteristic is Value. Value refers to the ability to extract useful information and insights from data that can support business decisions and improve Sales.

17. A Company stores large files in Hadoop Distributed File Systems. Which part of the Big Data ecosystem is used?

The Component used is HDFS (Hadoop Distributed File System). HDFS is a distributed Storage System that Stores large datasets across multiple Computers in a Hadoop cluster.

18. A Company wants to process large datasets using parallel Computation. Which Big Data tool can be used?

Apache Spark can be used. Spark processes large datasets using distributed and parallel computing, which make it Suitable for fast processing of Big Data.

19. An Organization wants to Collect live Sensor data from IoT devices. Which Big data ecosystem Component is Suitable?

Apache Kafka is Suitable for Collecting and handling real-time Streaming data from IoT devices. It can receive and transmit large amounts of Continuous Sensor data for further processing.

20. A Company wants to visualize big data results for decision-makers. Which tool or method can be applied?

The Company can use data visualization tools such as Tableau or Power BI. These tools Convert Complex data into charts, graphs, dashboards, and reports, making the results easier for decision-makers to understand and use.